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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR /IMPROVEMENT/ REAPPEARANCE EXAMINATION NOVEMBER 2025

FIRST SEMESTER
Discipline Specific Core Course (DSC) - MG1DSCECC102 - COMPUTER FUNDAMENTALS

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer **All** Questions

Each Question carries **1** mark

1. **A student is learning about the different layers of software in a computer and discovers that one layer is responsible for managing the computer's memory and processor.Which type of software is responsible for managing system resources and providing a platform for application software?** **[U] / [CO1]**

   System Software

2. **Name the cycle during which a machine language instruction is executed by the processor** [K] / [CO1]

   Machine cycle

3. **A computer performance report improvements after adding a memory component that stores copies of data frequently accessed by the CPU.What type of memory contributed to this performance boost by providing quicker access to critical data?** [U] / [CO1]

   Cache memory

4. **Write the type of computer that is used as a web server.** [K] / [CO1]

   Minicomputers

5. **Identify the mechanism used in optical storage devices to retrieve and store data.** [U] / [CO2]

Laser Light

6. **Identify the two types of SSDs that are classified according to their memory architecture.** [K] / [CO2]

Flash memory-based SSD and DRAM-based SSD

7. **List different types of impact printers.** [K] / [CO3]

Dot -matrix printer,Daisy Wheel printer,Drum Printer

8. **In which industry is MICR technology predominantly used?** [K] / [CO3]

Banking

9. **What file format might you choose when saving a photo from a digital camera to ensure a smaller file size?** [K] / [CO3]

Joint Photographic Experts Group

10. **Identify the device which exhibits the point,click and drag functions from the following: Trackball ,Joystick,Mouse and light pen.** [U] / [CO3]

Mouse

[10x1 = 10]

Part B

Short Answer Questions

Answer any **4** Questions

Each Question carries **5** marks

11. D**iscuss briefly the various characteristics of a computer.** [U] / [CO1]

Only two (2 marks)

Four or more than that (3 marks)

12. **Explain the importance of a bus in the computer system ,and list different types of buses commonly found in computer systems.** [U] / [CO1]

Bus definition (2 marks)

Types (3 marks)

13. **Describe how information is stored in a computer system's memory ,with a diagram**. [U] / [CO2]

Diagram (3 marks)

Explanation (2 marks)

14. **Describe the principle used in magneto storage systems for recording data**. [K] / [CO2]

Introduction (2 marks)

Working principle (3 marks)

15. **Discuss the concept of a plotter and explain the differences in design and function between flat-bed and drum plotters.** [U] / [CO3]

Concept of plotter (2 marks)

Differences between flat-bed and drum plotters (3 marks)

16. **Explain the categories of keys found on a typical keyboard.** [U] / [CO3]

Use of keyboard (2 marks)

Categories of keys (3 marks)

[4x5 = 20]

Part C

Essay Questions

Answer any **2** Questions

Each Question carries **10** marks

17. **Show and explain how communication between the processor and the I/O devices of a computer system is implemented. Write the steps for I/O write, including its block diagram.** [K] / [CO1]

Block diagram (5 marks)

Steps (5 marks)

18. **Illustrate the advantages of having a storage system in a computer system. List the different types of magnetic storage systems and provide examples of each.** [K] / [CO2]

Advantages of storage system (3 marks)

Types of magnetic storage systems (5 marks)

Examples (2 marks)

19. **Classify various scanning devices based on different scanning technologies, and describe each in detail with examples.** [U] / [CO3]

Classification of scanning device (3 marks)
Explanation (5 marks)
Examples (2 marks)

[2x10 = 20]